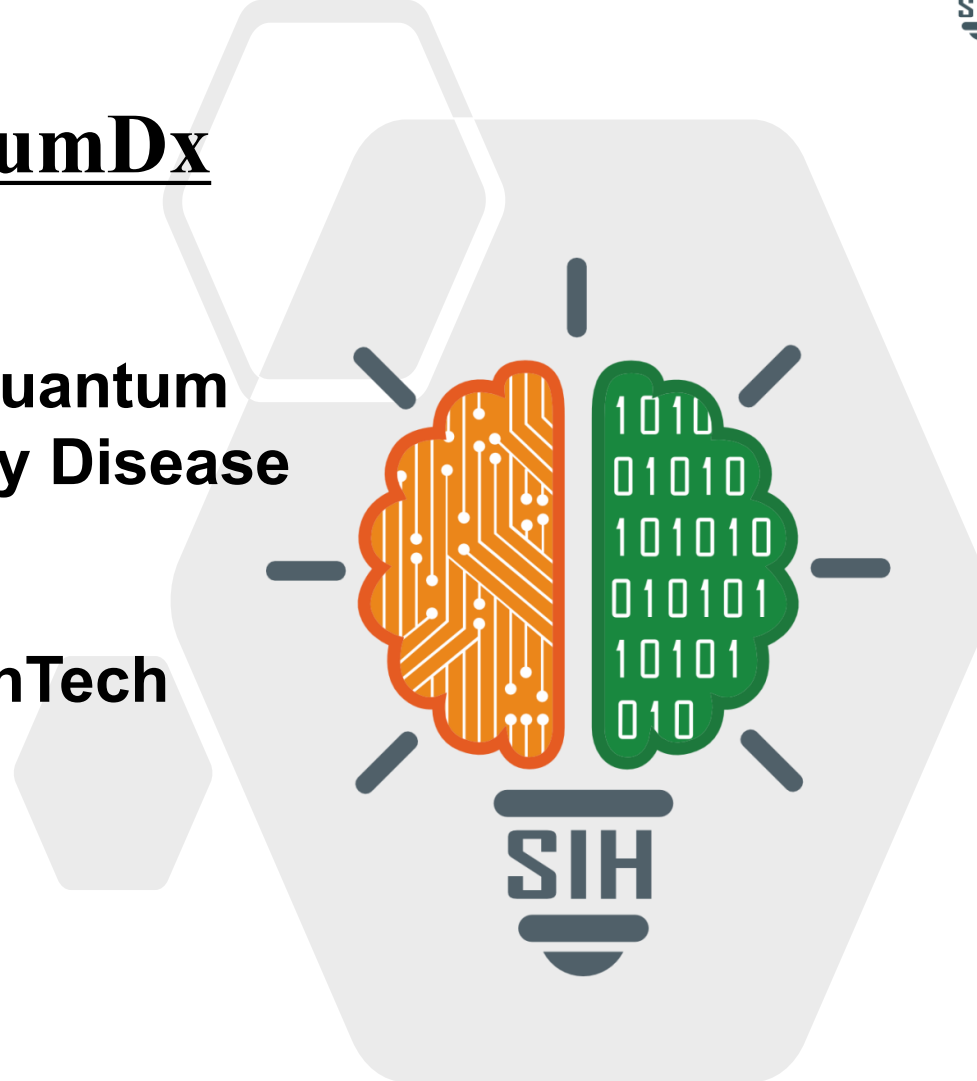


QuantumDx

- Problem Statement ID – SIH26139
- Problem Statement Title - Hybrid Quantum Machine Learning Platform for Early Disease Detection
- Theme - MedTech / BioTech / HealthTech
- PS Category - Software
- Team ID - 149574
- Team Name – Qunatum Bodha

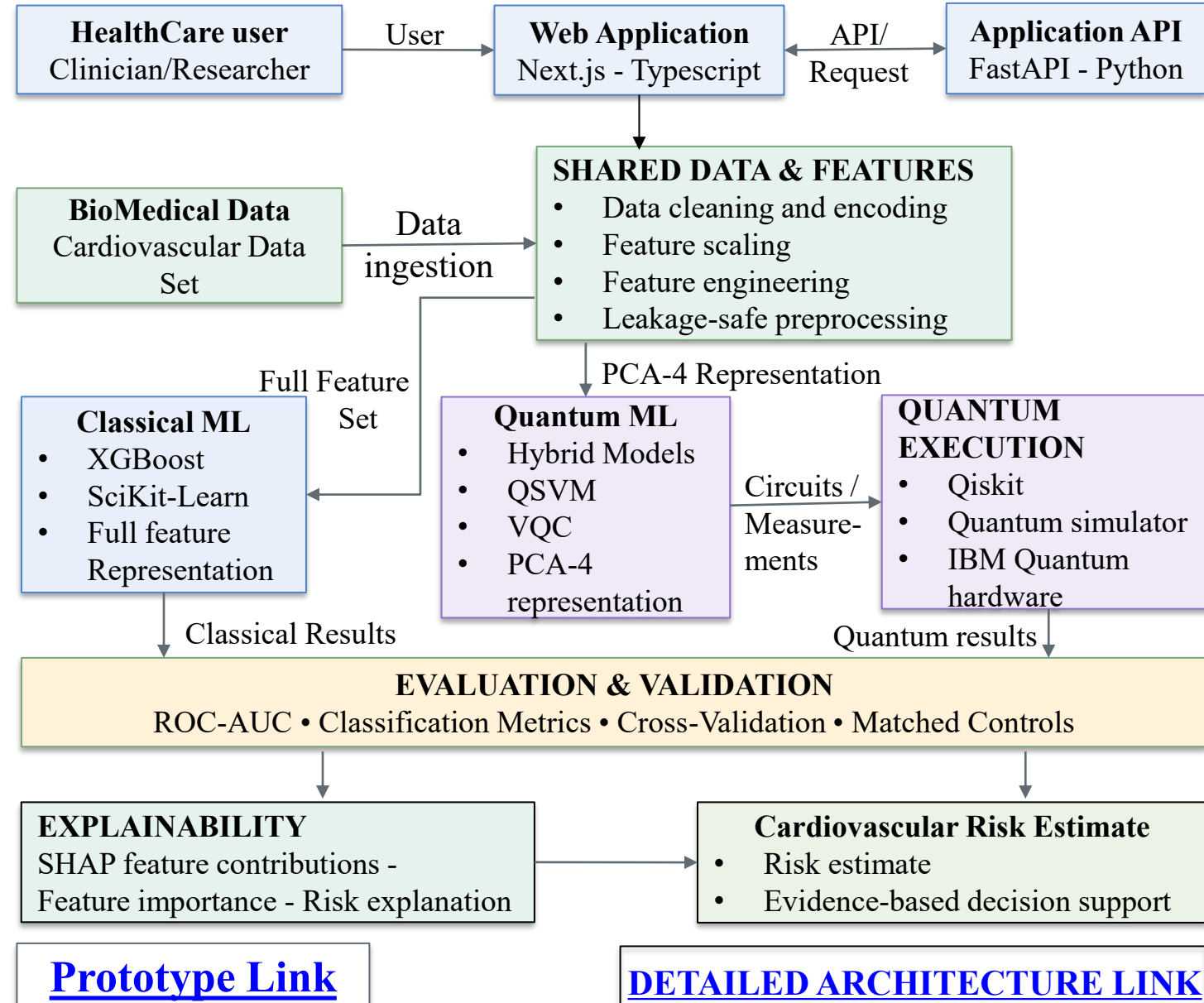


Proposed Solution :

Explainable Hybrid Quantum-Classical Framework for Early Cardiovascular Disease Risk Detection

“QuantumDx provides a leakage-safe biomedical ML pipeline that preprocesses cardiovascular data and evaluates classical, quantum and hybrid learning approaches under a common experimental framework.”

- **Leakage-Safe Data Pipeline :** Imputes, encodes and scales features, performs train-only feature selection and PCA, and generates separate classical and quantum-ready representations.
- **Hybrid Learning Pipeline :** Evaluates classical models, including XGBoost, alongside quantum models such as QSVM and VQC using a PCA-4 representation.
- **Quantum Execution:** Supports exact statevector simulation and real IBM Quantum hardware through a common backend system.
- **Explainable Risk Assessment:** Provides risk estimates, with SHAP-based feature explanations for classical predictions.



Technology Stack

Frontend : Next.js, React , Tailwind CSS.

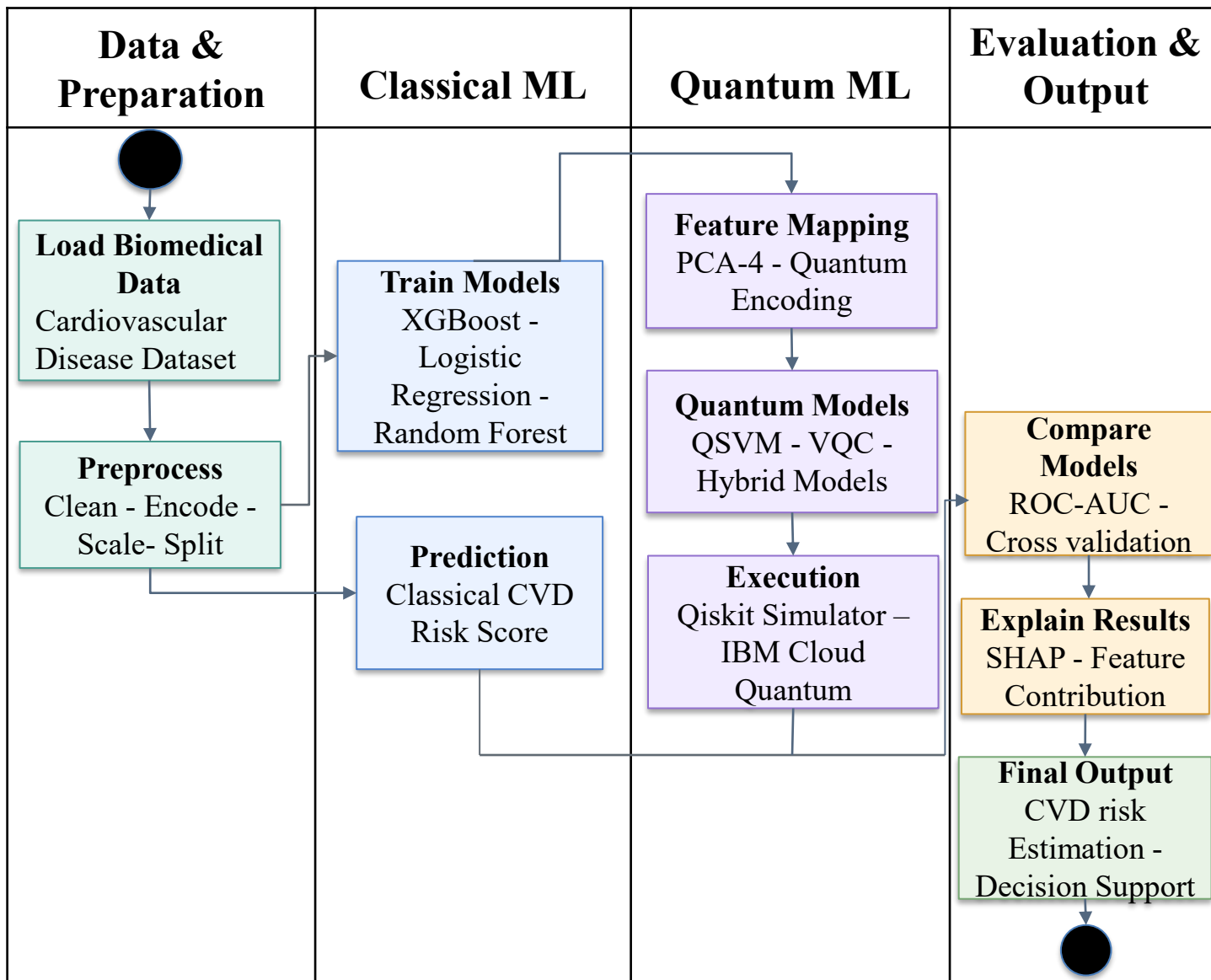
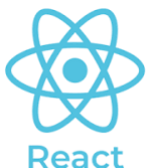
Backend : Python, FastAPI

Machine Learning : scikit-learn, XGBoost, NumPy, Pandas.

QUANTUM ML : Qiskit, Qiskit Machine Learning, QSVM, VQC, Parameterized Quantum Circuits, IBM Quantum.

DATA & MODEL PIPELINE : PCA, Feature Selection, SHAP

DEPLOYMENT / TOOLS : Git, GitHub, AWS.



[PROJECT WALKTHROUGH](#)

Complete solution walkthrough

[MANIM WALKTHROUGH](#)

Visual explanation of the system flow

FEASIBILITY:

- **Technology:** Python, XGBoost, Qiskit, FastAPI, Next.js
- **Data:** 70,000 raw records, 66,641 records in the full-scale experimental pool ([Dataset Link](#))
- **Implementation:** Classical ML, QSVM, VQC and SHAP evaluated, **Real time Quantum hardware** execution demonstrated through IBM cloud Quantum Hardware.

VIABILITY:

- **Performance:** XGBoost achieved **0.85669 ROC-AUC** on the full-scale dataset, quantum residual refinement reached **0.85649 ROC-AU**, closely matching the classical baseline.
- **Hybrid Validation:** Quantum refinement differed from XGBoost by only **-0.00020 ROC-AUC**.
- **Practical Use:** Risk prediction operates on classical computing without requiring quantum hardware.
- **Future Scope:** Re-evaluate quantum models as hardware and algorithms improve.

APPLICATION & RESEARCH POTENTIAL :

- **Hospitals & Clinics:** Cardiovascular risk-stratification and preliminary screening support.
- **Public Health:** Can complement large-scale NCD screening.
- **Research Institutions:** Benchmark classical, quantum and hybrid methods on biomedical datasets.

DATA SCALE

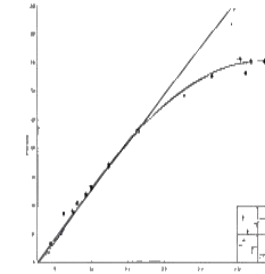
70K

raw records

full-scale experimental records



MODEL VALIDATION

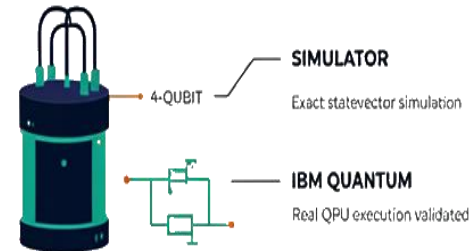


0.8567

ROC-AUC

XGBoost • full-scale benchmark
Matched with classical
• statistical variation

QUANTUM READINESS



DEPLOYMENT POTENTIAL



HOSPITALS



PUBLIC HEALTH



RESEARCH

SUPPORTING EVIDENCE

- “19.8M CVD deaths globally in 2022,” - [WHO, 2025](#)
- **India:** “220M estimated hypertension cases, only 12% controlled”. - [WHO India](#)
- **2025 QML review:** “Only 16/4,915 studies met rigorous criteria, no consistent quantum advantage was observed.” - [npj digital medicine](#)

Impacts

RISK ASSESSMENT SUPPORT

Estimates cardiovascular risk from available biomedical factors to support preliminary assessment.

EXPLAINABLE DECISION SUPPORT

Provides feature-level explanations to make model outputs more interpretable.

EVIDENCE-BASED QML EVALUATION

Enables controlled comparison of classical, quantum and hybrid approaches instead of assuming quantum advantage.



Benefits

HEALTHCARE

Supports preliminary cardiovascular risk stratification and screening prioritization.

PATIENTS

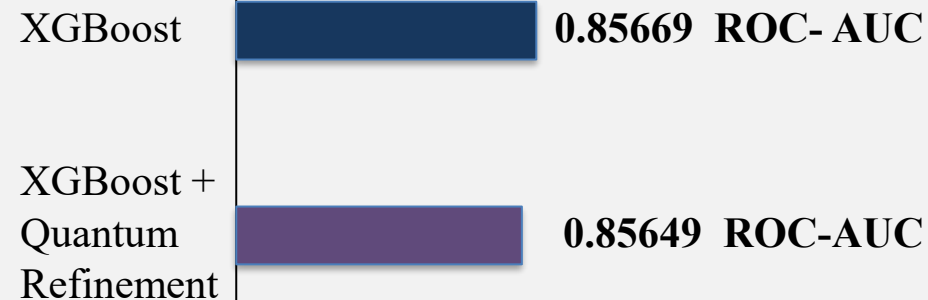
Provides understandable risk information to support appropriate follow-up.

RESEARCH

Provides a reproducible framework for benchmarking classical, quantum and hybrid ML on biomedical data.

PROJECT EVIDENCE

Model Performance (Full Scale ROC-AUC)



Data Scale

66,641
Full-Scale Experimental
Records

Quantum Validation

Near Term Quantum
Execution Successfully
Demonstrated Through
IBM Cloud Quantum

$\Delta = -0.00020$

**Quantum Refinement Closely Matched The
Classical Baseline**

S.No	Research Area	Research Finding	QuantumDx Decision	Result/ Evidence	Reference
1.	XGBoost	Gradient-boosted decision trees are effective and scalable for structured/tabular prediction.	Use XGBoost as the primary classical benchmark before evaluating quantum models.	0.85669 ROC-AUC on the full-scale dataset.	Chen & Guestrin, <i>XGBoost: A Scalable Tree Boosting System</i> , KDD 2016. DOI
2.	Quantum Kernels / QSVM	Quantum feature maps can create quantum-enhanced feature spaces for kernel-based classification.	Evaluate PCA-reduced cardiovascular features using quantum kernels and QSVM.	QSVM was benchmarked against classical models and matched non-quantum controls.	Havlíček et al., <i>Supervised Learning with Quantum-Enhanced Feature Spaces</i> , Nature, 2019. DOI
3.	Principal Component Analysis	PCA transforms correlated variables into a smaller set of principal components.	Apply PCA before quantum encoding to reduce the input dimension for quantum circuits.	4-component representation used for quantum experiments.	Jolliffe, <i>Principal Component Analysis</i> , Springer, 2002. DOI
4.	SHAP	SHAP provides feature-level attribution to explain individual model predictions.	Include explainability instead of returning only a cardiovascular risk score.	Feature-level contribution analysis incorporated into the prediction workflow.	SHAP, NeurIPS 2017 DOI
5.	QML in Digital Health	Recent evidence does not establish consistent empirical utility of QML over classical methods in realistic digital-health settings.	Do not assume quantum superiority. Use matched classical controls and statistical comparison.	No measurable contributions from quantum refinement was demonstrated under the tested conditions.	Gupta et al., npj Digital Medicine 2025 DOI